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Introduction to the Platform for Windows Programmers

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Objectives

- Palm Computing Programming vs. Windows Programming
- Moving a Windows Application to the Handheld
- Building Palm Expertise in Your Organization

Palm Computing Programming vs. Windows Programming

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- What is a Palm Computing® platform program?
- Development tools
- Platform issues

What Is a Palm Computing® Platform Program?

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- Code, resources and data
- PRCs and PDBs
- Program storage and execution

Development Tools

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- Programming languages
- C++: what's available, what's not
- Development environment choices
- Run-time library support

Platform Issues

- Memory (or lack thereof)
- Working with a small display
- File system
- No multi-threading or tasking

Moving a Windows Application to the Handheld

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- Application partitioning
- Less is more!
- User interface design
- Memory considerations
- Plan for debugging
- Create components where applicable
- Communicating with the outside world

Application Partitioning

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- Logical separation of application
- Palm™ application
- Conduit
- Host or desktop application

Less Is More!

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- Do not cram a Windows application's feature set onto the handheld
- Follow the 80-20 rule
- Aim for a small set of well designed, easy-to-use features

User Interface Design

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- De-emphasize user input
- Favor the view-only user
- Provide selection from limited choices vs. free-form entry
- Follow guidelines and standards
- Plan for the finger as well as the stylus!

Memory Considerations

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- Common misconception: Total memory does not mean available heap memory!
- OS 2.0 offered developers only 12K on the Professional, assuming TCP/IP
- Things have gotten better, but are still tight:
  - Approximately 36K available on OS 3.0-3.3, regardless of installed RAM
- Approximately 2K stack

Memory Considerations (Cont.)

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- #1 newbie mistake: loading lists of data into memory—don't do it!
- Read and edit data in-place
- Handles and pointers
- Don't lock down memory
- Make use of temporary storage

Plan for Debugging

- You do not want to do a lot of it
- POSE is an awesome tool—use it!
- POSE is great, but remember that your users will be using devices!
  - Do not wait a long time to test your app on a real device
  - Do not rely heavily on the same memory availability that you see under POSE
- Strategies for debugging
- Develop and test under Windows

Create Components Where Applicable

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- Static and shared libraries
- Shared libraries vs. DLLs
- Segments and large applications

Communicating with the Outside World

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- Serial
- Infrared
- Wireless
- Networking
- Printing

Building Palm Expertise in Your Organization

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- Bootstrap with outside expertise
- Establish gurus
- Don’t attempt a time-critical project first
- Obtain all the reference materials you can
- Palm resources and forums

Reference Materials

- SDK docs (PDF and paper)
- Books: There are now lots of resources for you to choose from!
  - Sams' Palm Programming: The Authoritative Solution
  - Sams' Teach Yourself Palm Programming in 24 Hours
  - Palm Programming: The Developers Guide
  - Palm OS Programming for Dummies
  - Palm Programming from the Ground Up
  - Palm Database Programming

Palm Resources

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- Developer Forums
- www.palm.com/devzone
- FAQs and knowledge base
- Provider Pavilion
- Developer Support

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Q & A

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